



“Fasten your seatbelts, it's going to be a bumpy night.” (All About Eve, 1950)

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Purpose

This document belongs to ICT2113 module of the SIT cluster of the same acronym (ICT). Its purpose is to provide all learning and instructions necessary for the [successful conduct of the labtorials](#). To meet this mandate, this document provides details for the (a) class and team projects, (b) labtorial lesson plans, (c) deliverables, and (d) assessment. As well, it [uses and supplements material](#) presented in the learning sessions (LS), including the teaching and learning protocol, module master plan, and effective writing of requirements.

All events, deliverables, and assessments belong to the ‘Reqs now & near future class project’ for our module, hereafter referred to as [“the class project”](#). Undertaking the class project constitutes the majority of marks (just under two thirds) earned by the students for

our module. Therefore, students are required to read through this document however many times as it takes to understand the standards of performance expected by the teacher. If there is anything contained herein that is not clear nor cogent, the discovering student is encouraged to [seek supplemental explanation](#) from the teacher (author).

Project: Class vs team

The *class* project obliges the students to experience being in the roles of [customer](#) and [vendor](#) as they conduct the work comprised in [three requirements capture methodologies \(RCM\)](#), namely [Requirement Engineering \(RE\)](#), [User stories](#), and [UseCase 2.0](#). These three RCM are chosen to provide the students with a reasonable appreciation of the field of ‘requirements capture now and in the near future’. Requirements capture is known to be the most impactful and problematic stage in the development of new and not yet existing software, so this is a very important subject for the students to learn. The two roles necessitate a [significant change of perspective and responsibilities](#). It is anticipated that this will constitute a challenge for the students, but it is necessary to put them through this because this is a real challenge they will face in their software engineering careers. Finally, the context/ target of the work is a sample software development project, hereafter referred to as “[the team project](#)”; ergo, there are [three team](#) projects — corresponding one-to-one with the three RCMs — within the [one class](#) project.

Teaching and learning protocol

As been revealed in the Prologue section of the first learning session (LS), there is no external switch to make a student learn. To be sure, learning is an [internal and largely voluntary](#) process, in other words, the student must [want](#) to learn; only then is there a [potential](#) that learning will occur. Research has shown overwhelmingly that [motivation and engagement](#) are the keys to [successful](#) learning (Finn, 1989; Jones, 2008; Krause & Armitage, 2014). These two states of being are multi-dimensional because they have [psychological and emotional](#) components. Triggers for these states include students feeling liked, respected, and valued by their teacher and peers, or having a sense of belonging to their group. Also, they are independently manifested. Motivated students are usually engaged, but not always; those who are shy may not be engaged in a socio-cultural environment like a physical classroom. Students who are engaged but not motivated may have other influences, such of paper chasing or relationships with other students.

There are three additional aspects to engagement: [relatedness, competency, and autonomy](#). [Relatedness](#) — is the activity requiring the student’s engagement commensurate with his/hers chosen area of subject or interest? In a lot of cases, yes, the student has already a

good idea of what they want as a career and what are the important skills they need to learn. **Competence** — do the students feel that they have developed a degree of self-efficacy, that their actions are worthwhile and cogent? If the students experience a strong sense of capability in the learning environment, they will be far less reticent in engaging in challenging tasks and activities. Finally, **autonomy** — can the student have some level of control over their work? Maybe not in the learning outcomes of assignments and their deadlines, but perhaps in the content and exercising of imagination and creativity in the execution of those activities. This makes the precise delineation of engagement as, the ‘right kind of engagement’, a.k.a., following best practice. So, the keys to **unlocking** a student’s learning success are **motivation and following best practice** (Martin, 2009; Jones 2023).

While learning is **not** externally-triggered, motivation and best practice **can be, by teachers and peers alike**. **Effective triggers** include (a) providing of regular and informed **feedback**, (b) encouraging students to **evaluate their own learning** (not marks) progress/be introspective, (c) **sharing** what the cohort is doing right, (d) providing **examples** intended to unlock the challenging concepts, (e) recognizing student’s **proactivity** in any task, (f) establishing group **collaborative** learning opportunities, and (g) promoting active participation by students in class (Schlechty, 2001; Jordan, 2012; Reschly & Christenson, 2012).

In summary, research shows that facilitation of a student’s learning is the aggregate of the quality, enjoyment, and mutually supportive nature of relationships and interactions with teachers and peers (Jones, 2008; Reyes, Brackett, Rivers, White & Salovey, 2012). To accomplish this, the teacher must succeed at providing the students with, (h) **emotional support** by being subtle, thoughtful, receptive, sincere and conscious of the interests and needs of their students, (i) **organisational support** by creating orderly classroom environments, where expectations are clearly stated and students develop more productive learning habits, and (j) **instructional support** by providing clear content, guidance, and feedback to the students (Pianta & Hamre, 2009). Commensurately, the student must adopt this compatible learning protocol:

1. **Decide** to proceed either as a teacher-driven learner, or self-directed one! If former, **listen & read carefully**, and **accept & regurgitate**; and if latter, **plan, acquire pers library, read critically, find what is missing**, and **self-test new capability** (Jones, 2023).
2. Adopt learning **strategy** for module: surface, deep, or strategic. (Prosser & Trigwell, 1999.)
3. Establish & maintain learning **rhythm**: what time of day, what place, how many days per week. (Jones, 2008)
4. For each topic, decide learning **goals** (in order of difficulty): know **gist & terms**, understand **fundamental concepts**, **apply** learned concepts to **solve problems**,

analyze how to **improve** yourself, evaluate solutions of **others**, and create **other ways** to achieve similar/better result. (Bloom's taxonomy, revised 2001.)

- Above all, **listen & adapt** accordingly, be both **creative & critical** thinker, make **every choice count**, and **communicate your uncertainties** to collaborators and teacher; choose wisely. (Robinson, 2016)

Important! Teaching is **dialectic**, meaning it is via **dialog** and **reasoned argument**.

Module master plan

Wk	List of learning sessions per exact filename	Category of LS
1	Reqs Philosophy, Process, Methodology & Modeling	gain attention (1st event)
2	Capturing 'Big Reqs Up Front' – Reqs Engr (RE)	reqrs now
3	Eliciting Authentic & Right Reqs by Customer	
4	Knowing Right Reqs Models – UseCase,ERD,DD, etal	modelling
5	Analysing PS Reqs By Vendor Proficiently wBCIC	reqrs now
6	Specifying Reqs in SRS wQual+iRTM+ChngProtocol	
7	Recess week	
8	Validating – QC SRS by Customer-Vendor & QA RE	reqrs now
9	Trending Sw Reqs in SE Asia SME – Industry G-Lec	guest lecture
10	Overcoming Deficiencies of RE – UserStory is Answer	reqrs near future
11	UseCase2.0 Heavyweight RCM Updated to Lightweight	
12	UML Giant No More – Less Relevant in Lightweight	modelling
13	Taking Your Learning From Uni to SWE Industry	retain & transfer (9th event)

Fig 1. Schedule of LS for the module.

Wk	Category of LS	Lab session	Team assmt (upon post)	Individual assmt (during)
1	gain attention	labtorial Proj, PD note, discn NEID	←	{ quiz + n during LS & lab }
2	reqs now	labtorial RE, lrn protocol	←	{ quiz during lab }
3		labtorial Eliciting	←	
4	modelling	posting doc	{ post PS spec }	
5	reqs now	labtorial Analysing	←	{ prez during lab }
6		present doc	{ post Analysis rpt }	{ quiz during week }
7	Recess week		←	
8	reqs now	labtorial Specifying	←	
9	guest lecture	posting doc	{ post SRS spec }	
10	reqs near future	labtorial Validating	←	{ prez during lab }
11		present doc	{ post Validation rpt }	
12	modelling	labtorial US & UC2.0	←	{ quiz during week }
13	retain & transfer	posting doc	{ post UC & US rpt }	

Fig 2. Relation of LS to labtorials to assessments in the module.

Figures 1 & 2 are repeats of the [master plan](#) images shown in the first LS, so one doesn't have to flip back to the slides to refresh on these two crucial references.

Figure 1 lists [all the LS](#) that are contained in the module and the [sequence](#) in which they are presented. Requirements "now" belong to the RE RCM; "near future" is User Story and UseCase 2.0 RCMs. "Modelling" is requirements models and UML. "Guest lecture" is sharing by a consultant of some [trends](#) in requirements capture in South-East Asia. Finally, "1st" and "9th" events bookend the module in accordance with [instructional design best practice](#) (NEID by Gagné) shared in the first LS.

Figure 2 shows the relation between the LS, labtorials, and assessments. Crucially, LS always precede their associated labtorials. Most labtorials are conducted over [two weeks](#), with [teaching](#) in the first week and [finalizing work](#) in the second. All teaching in labtorials is [dialectic](#). The outcome of each labtorial pair is a [document deliverable](#). There is an additional outcome of individual presentations in two labtorials.

Weekly labtorial plans

This is the schedule of lesson plans for all the weekly labtorial periods in the module. Every entry shows the [week number](#), teacher's [title of the labtorial](#), and its [plan of conduct](#). Note that Week 7 is the [recess break](#) in the module, and there is no labtorial period.

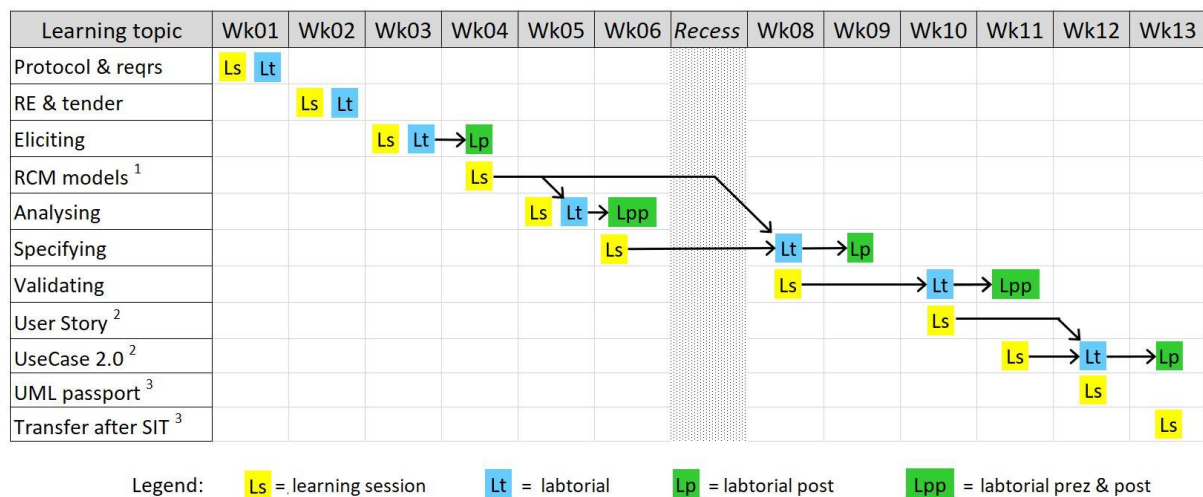


Fig 3. Schedule of labtorials linked to LS in the module.

Notes:

- ¹ "RCM models" LS has [no](#) labtorial. Instead, its content can be used in Analysing and Specifying document deliverables.
- ² "User Story" and "UseCase 2.0" LS have one labtorial each; the two are [combined](#) to generate one deliverable document.
- ³ "UML passport" and "Transfer after SIT" LS have [no](#) labtorials.

Universal precepts for all labtorial periods

1. Students must immerse themselves fully into the workings of each labtorial period's stated "**Role**". To demonstrate that this is being done in earnest, the deliverables produced for the team project must have the appropriate corporate identity items that a **real organization in today's commercial world would have**. The minimum corporate identity items are listed in Activity #2.
2. Students must apply in the labtorial periods, not only the **facts** and **rationale** of every stated LS, but also prior **knowledge and experience**, **inference**, **creativity**, and a dash of **innovation**. (Note: If this module's learning material seems to clash with another module, please see the teacher for clarification.) Particularly, this must be the main practice of the students in the **dialectic sessions**, when the salient points of each LS are revisited, and the students have an opportunity to interrogate those parts of the LS material that are not clear.

Week 1 Prepare team project

Role: Customer.

Synopsis: You are waiting in the conference room of the headquarters of your employer company, along with seven other people. (A few you know already, **others not so**.) Your reporting officer explained that the CIO was calling the meeting to brief the team that will **oversee the development of the latest software** for the company, and that you are nominated to be your department's representative on that team. You knew of the project, but not any of its details. Just as you were about to find out if any of the other persons in the room knows more about the project, the CIO strides in and calls the meeting to order.

Knowledge base: LS01 Reqs Philosophy, Process, Methodology & Modeling.

Activities:

1. Students in each labtorial group prepare for *class* project. (45 mins.)
 - Students **form** teams, preferred size is **five** members. Anything less or more will be exceptional and adjudicated case-by-case.
 - Team appoints one of its members to be its **Point of Contact (PoC)**.
 - Team adopts a **team name** to which all marking will be awarded. The name is PoC's **last four digits** of his/her **matric number** plus **first letter of surname**, e.g., 1234A.

- Team selects a **Product Description (PD)** that corresponds to the assignment of the *team* project. Each PD is assigned to **only one** team **A**. See Appendix A for further clarification of the rationale of the PD for the *team* project.
 - After PD assignment is approved, the PoC **immediately downloads PD** and **distributes** it to all members. (The teacher will remove it from the LMS.)
 - Lastly, the PoC **emails the teacher** with (a) list of members, (b) team name, (c) approved PD, and (d) declaration of distribution of PD to all members, upon collecting all the info.
2. Students individually analyse their team's approved PD. (30 mins.)
- Team members disperse amongst tables in venue so that only one from each team is at every table **B**. (Remember this table for next week.)
 - Students independently conduct a **survey** of the **suitability of their respective PD's contents**, e.g., is it clear to me? what don't I understand? is it doable? is this feasible (not pie in the sky)? what is missing that that my experience says it should be present? is this software for our use or for selling? and, what kind of technology does this involve?
 - Students **record** their ideas on a '**Note**' (individual)) and post in appropriate **dropbox**.
3. Teams devise **pretend identity** and **backstory** of **two** employer companies, one being **customer** and other being **vendor**. (15 mins.)
- Team brainstorms the company's **names**, **logos**, **address'**, **letterheads'**, **commercial sectors**, their **line of business**, and their **recent history** as a customer or vendor in the industry suggested by the PD.
 - PoC record details of both companies on a '**Note**' (for team) and post in appropriate **dropbox**.
4. Students engage in a **dialectic session**, as a cohort of individuals (not team members), on the learning potential of **NEID**. (30 mins.)
- Teacher asks about **first three** events {gain attention, objectives or outcomes, old or prior knowledge}. How they work? why they are in order? what do you personally expect to gain? what are learner's responsibilities? how do you know you are successful in the events.
 - Teacher asks same things about events **four to eight** {do in chunks, tips or learning aids, elicit practice, give advice or feedback, confirm learning}.
 - Teacher asks same things about **ninth** event {hold onto learning}.
 - Student records the discussion's salient points for review and recall.

Deliverables: There are **three**: a) "**Prerequisite quiz appraisal**" filled in during LS#1, b) "**Customer and vendor company details**", and c) "**PD survey**"; last two are written in the

labtorial Activities [#2 & 3](#). Post all into the LMS Dropbox folders with the appropriate file names and by the fixed deadline (see *Deliverable details* section). Teacher's comments will be annotated to submissions by the following week.

Notes:

- A** During the labtorial, each team takes a queue number and announces their preference of PD in FIFO; teacher ensures that no two teams have the same PD. If the PD is not yet assigned to another team, the PD is immediately assigned to the requesting team. If already assigned, the team must immediately announce an alternate preference; if it can't do this on the spot, it must leave the queue and return later. Finally, if a team forms before the labtorial and emails its PD preference to the teacher, it will be processed in FIFO order of receipt either before or at the start of the labtorial.
- B** All but one of the team members stand up away from their tables. Team by team, the members seat themselves one at a time at successive tables until all members are dispersed one to each table.

Week 2 Overview of RE RCM

Role: Customer.

Synopsis: Your team seems to be working well together. It has already studied the PD written by the senior management; unfortunately, it has a lot of holes that are generating more questions than answers. In addition, your team still doesn't know the timeline for the tender process to select the vendor, and what requirements capture methodology (RCM) will they be following. So, your team is back in the headquarters conference room, waiting for a briefing from COO and CFO that will hopefully provide all the answers your team seeks. Finally, the two C-staff and their senior managers arrive.

Knowledge base: LS02 Capturing 'Big Reqs Up Front' – Reqs Engr (RE), LS01 Extending Reqs, Process, Methodology & Modeling, and LS00B Learning & Teaching Protocol.

Activities:

5. Teams collectively engage in [dialectic session](#) on the following contemplations posed by the teacher with respect to LS02 & 01. (30 mins.)
 - Why does SDLC matter to you, and what do you do when according to the process, you (as customer) not supposed to be participating?
 - What are requirements capture and its two kinds of methodologies?
 - What is your primary concern (as customer) about working with heavyweight?
 - What is your team's responsibilities in tender process?
 - Where in the tender process is the risk of setting the project onto a path of certain failure the highest, and what can you do about it?
 - Student takes personal session notes.

6. Teams individually engage in [brief feedback session](#) with teacher on PD issues. The teacher spends approx. 10 mins with each team. If not enough, then PoC encouraged to request an [additional session](#), preferably remote on zoom. (60 mins.)
7. Students do [quiz](#) on learning & teaching protocols from LS00B & LS01. (30 mins)
 - Team members disperse amongst tables in venue so that only one from each team is at every table [B](#). (Go to table from last week.)
 - Students open and complete quiz in LMS within preset time limit.
 - (Students download Appraisal Sheet from LMS [C](#).) Teacher presents solutions. Students appraise their responses, and fill-out then total appropriate table in template.
 - Student immediately posts his/her note into LMS.
8. Students engage in brief [dialectic session](#) on their preparation for and conduct during learning sessions. (15 mins.)

Deliverables: There is [one](#): “[Learn & teach protocol appraisal](#)”. Post it into the LMS Dropbox folder with the appropriate file name and by the fixed deadline (see *Deliverable details* section). Teacher’s comments will be annotated to submissions by the following week.

Notes

- [C](#) Students must do this [before](#) coming to labtorial.

Week 3 Eliciting method | Preliminaries

Role: Customer.

Synopsis: After the briefing last week, your team now has the big picture of the project, including the RCM. (It’s [RE](#), a [heavyweight](#) methodology.) Your first major mandate is to deliver the PS to the CFO so that the tender process can be activated; in other words, entering the Eliciting method, first in RE. To enable meetings with stakeholders and the eventual vendor, your team has been assigned a dedicated meeting space large enough for all the members plus equal number of guests, and a dedicated cloud storage account with a walled-off sandbox for prototypes, software testing, and AV; the former for the next eight months and the latter for as long as it is needed (could be for years). Now your team is waiting for a briefing by HR about the stakeholders, and IT staff about using the cloud account for uploading code.

Knowledge base: LS03 Eliciting Authentic & Right Reqs by Customer. See [Appendix B](#) for further clarification of the effective writing of new requirements.

Activities:

9. Teams collectively engage in [dialectic session](#) on the following contemplations posed by the teacher with respect to LS03. (30-45 mins.)
 - What is method's goal?
 - Why is the method not consistent, i.e., not done in the same way by different teams, or not able to deliver similar results over and over again (what are idiosyncracies, variables, obstacles, uncertainties)?
 - What major aspects of elicitation should method's plan address (relate to goal)?
 - What is reasonable expectation of elicited requirements (experience level, what is most important characteristic)?
 - How and from whom are four sections of PS best written?
 - Why is it crucial that elicitor has experience in the requirements biz context?
 - If the method is not consistent, why then should PS be templated (granularity, necessity of understanding)?
 - Do project members bias their eliciting when they are stakeholders?
10. Teams [apply](#) concepts, hints, examples and ideas from knowledge base and dialectic session to the [undertaking](#) of the document deliverable. (75-90 mins.)

Week 4 Eliciting method | Finalizing

Role: Customer.

Synopsis: The Eliciting method is fully underway. After the HR and IT briefings, your team quickly decided upon the [Eliciting Realization type\(s\)](#) and [elicitation techniques](#) that it would use for capturing the new software's capability and condition information from the stakeholders on the list provided by HR. Now that the stakeholders' sessions are completed, and all the information secured, your team is writing the PS and touching up the necessary [models](#). (Note: Your team has assistance on models at any labtorial from this week onwards.) Your team is confident that it will meet the agreed deadline for submitting the PS to the CFO for the forthcoming tender meeting.

Knowledge base: LS03 Eliciting Authentic & Right Reqs by Customer.

Activities:

11. Students pose their questions to teacher concerning any preliminary work on their Eliciting method, either on behalf of their teams or as individuals. (~30 mins.)
12. Teams continue their [undertaking](#) of the deliverable document. (~90 mins.)

Deliverables: There is **one**: “**Eliciting PS**”. Post it into LMS Dropbox folder with the appropriate file name and by the fixed deadline (see *Deliverable details* section). Teacher’s assessment will be emailed to PoCs after the following week.

Week 5 Analyzing method | Preliminaries

Role: Vendor.

Synopsis: Your employing company is all abuzz over winning the bid for a large software development project. The Project Manager (PM) assigned to lead the effort has put the development team together. Strangely enough, you (a rookie employee) along with a few others (whom you don’t know) have been chosen to execute the RE Analysis and Specifying Methods. You are quite anxious to do a good job, but worried about making a mistake. Afterall, Analysis and Specifying are crucial methods in any software development project, so any mistake by you could have catastrophic consequences for the whole project. Two days ago, the **PS** from the customer was posted into your email inbox, and you’ve read it three times over in preparation for today’s first briefing with the PM. What is the PM like? What is the customer like? Are there team members with more experience that you can leverage for support if needed? What is this **BCIC sequence** you’ve heard so much about? You make it to the briefing room, just in time to have the PM hurry you to your seat. Almost growling, the PM announces, “Next time we meet, you will ALL be on time!”

Knowledge base: LS05 Analysing PS Reqs by Vendor Proficiently wBCIC, and LS04 Knowing Right Reqs Models – UseCase,ERD,DD et al.

Activities:

13. Teams collectively engage in **dialectic session** on the following contemplations posed by the teacher with respect to LS05 & 04. (30-45 mins.)
 - Why do you think the terminology “Analysing requirements” is so misapplied?
 - Why should the vendor prepare separate and presumably implementation-suitable requirements instead of directly editing the PS and then publishing a new version of it?
 - Why is it crucial that the separate and presumably implementation-suitable requirements are derived from a transmutation of the customer’s PS?
 - How can the vendor proceed if it doesn’t have an Account Manager for the customer?
 - When in the Analysing method is having experience in the requirements business context most important?
 - What are the benefits of loading the ready-to-specify requirements into a repository before populating the SRS (codifying, granularity of requirements)?

- What actions can the vendor take to mitigate some of the risks that are present in Analysing method.
14. Teams **apply** concepts, hints, examples and ideas from knowledge base and dialectic session to the **undertaking** of the document deliverable. (75-90 mins.)

Week 6 Analyzing method | Presentation & finalizing

Role: Vendor.

Synopsis: It has been several very anxious weeks for your team working with the customer's stakeholders. It is not they've been unwilling to work with you. Instead, it's more that they are so non-committal about some of the omissions and vagaries of the requirements. You were briefed about the risk of this happening, as it is difficult for the customer to imagine a new technology they have never used before. Equally difficult is you trying to put yourself in the customer's shoes to know what would be the most effective step forward in the new technology. Nevertheless, your team has gotten most of the omissions and vagaries resolved, and now is coming up with the **interpretations and categorizing**. Unexpectedly, your team receives a Whatsapp message from the PM to provide him with a presentation in 24 hours on how far along you are with the Analysing Method. You do an all-nighter to get all the statistical data ready on the errors and resolutions you've reached onto slides, along with an accompanying narrative on what happened in the meetings with the stakeholders. After couple of cups of "jet fuel" — your canteen has terrible coffee — you head over to his office.

Knowledge base: LS05 Analysing PS Reqs by Vendor Proficiently wBCIC, and LS04 Knowing Right Reqs Models – UseCase,ERD,DD et al.

Activities:

15. Teams **present** their Analysing method reports. Teams in the audience may utilize whatever inspiration they find from the presenting team in their own document deliverables. (~120 mins.)

Deliverables: There are **two**: (a) "**Analysing Rpt**", and (b) "**Analysing Prez**". Post both into LMS Dropbox folder with the appropriate file names and by the fixed deadline (see *Deliverable details* section). Teacher's assessment will be emailed to PoCs ASAP the following week.

Week 8 Specifying method | Preliminaries

Role: Vendor.

Synopsis: Your report presentation to the PM went well. He seemed satisfied that the Analysing was on track. Anyway, your team was able to complete Analysing in good time, with all of the transmutations of the PS uploaded into the repository. Now, your team is

starting on the Specifying method. Your team has [selected the template](#) for the SRS, and has split up the work of filling it out, evenly between the members. Also, your team will be putting in the company's authorized [quality model](#). Of course, you all know that your greatest challenge is coming up with a [codification system](#) that is adaptive for the all the special requirements, but simple to apply across the template's sections plus conducive to aggregation in the [RTM](#). Your team is in the middle of brainstorming to figure all of that out.

Knowledge base: LS06 Specifying Reqs in SRS wQual+iRTM+ChngProtocol, and LS04 Knowing Right Reqs Models – UseCase,ERD,DD et al.

Activities:

16. Teams collectively engage in [dialectic session](#) on the following contemplations posed by the teacher with respect to LS06 & 04. (30-45 mins.)
 - Do you choose the SRS template because, it shows the content you have determined is needed, or tells you what content you should have?
 - Why is it unlikely that the vendor's quality model for the SRS would clash with the customer's (pri not what, relational)?
 - How do you establish or put forth in the SRS, the usage of the attributes of the quality model?
 - How much info should be put into the enumerative scheme of the RTM's codification?
 - How does mitigating the change of the SRS actually help in reducing errors in the rest of the SDLC after requirements?
17. Teams [apply](#) concepts, hints, examples and ideas from knowledge base and dialectic session to the [undertaking](#) of the document deliverable. (75-90 mins.)

Week 9 Specifying method | Finalizing

Role: Vendor.

Synopsis: So far, so good. Your team has worked out a suitable codification system for the SRS template, and the RTM is being completed according to schedule. Now, the final challenge is working out the [change mitigation strategy](#) for the SRS going through the main development stages. Your team has come to realize that change is inevitable, not because of poor process or work, but because stakeholders and developers alike want to develop the best software ever. Against that are forces beyond control, such as market shifts, new technology, new people; the heavyweight SCM cannot protect the development team from these happening. So, your team will derive a strategy that will give the development team the tools to make it through any such potentially destructive changes. Once completed, the SRS will be sent to the PM for his authentication.

Knowledge base: LS06 Specifying Reqs in SRS wQual+iRTM+ChngProtocol, and LS04 Knowing Right Reqs Models – UseCase,ERD,DD et al.

Activities:

18. Students pose their questions to teacher concerning any preliminary work on their Specifying method, either on behalf of their teams or as individuals. (~30 mins.)
19. Teams continue their [undertaking](#) of the document deliverable. (~90 mins.)

Deliverables: There is [one](#): “[Specifying SRS](#)”. Post it into LMS Dropbox folder with the appropriate file name and by the fixed deadline (see *Deliverable details* section).

Teacher’s assessment will be emailed to PoCs ASAP the following week.

Week 10 Validating method | Preliminaries

Role: QC vendor and QA vendor.

Synopsis: The PM has authenticated the SRS. Now, in accordance with the company’s project management policy, the PM invited two independent auditor firms to confirm that there has been no compromise of company policy and quality in both the [SRS](#) and the RE [processes](#) that produced the SRS. (The audit of the SRS document is called [Quality Control or QC](#); for the process, it is called [Quality Assurance or QA](#).) The PM will handle the QC audit on behalf of our company, and the COO will do the same for the QA. Your team will provide the QA auditor with all the documents from the Analysing and Specifying Methods — plans, journal entries, notes, claims, drawings — they need from our company. (Note: The QA auditor will also confirm the [integrity of the QC audit](#). To that end, the QA auditor works directly with the QC team for the necessary information.)

Knowledge base: LS08 Validating - QC of SRS by Customer-Vendor & QA RE.

Activities:

20. Teams collectively engage in [dialectic session](#) on the following contemplations posed by the teacher with respect to LS #8. (30-45 mins.)
 - Analysing has quality attributes of requirements, Specifying has company quality model, and Validating has quality management: Why so much quality?
 - Why should errors targeted in Validating QC be different than in Analysing and Specifying?
 - Who is ultimately responsible for hiring the independent auditors for QC and QA?
 - If the independent auditor finds a QC problem, what can and should be done?
 - If independent auditor finds a QA problem, what can and should be done?
21. Teams [apply](#) concepts, hints, examples and ideas from knowledge base and dialectic session to the [undertaking](#) of the document deliverable. (75-90 mins.)

Week 11 Validating method | Finalizing

Role: QC vendor and QA vendor.

Synopsis: The two independent auditors have completed their work, and today they are giving their [respective reports](#) on the [quality plus policy adherence of the SRS and the RE processes involved in the production of the SRS](#). Your team knows they did the job in strict adherence to the orders and regulations. Still, you can't help but feel a little apprehensive as you're sitting in the meeting room waiting for the QC auditor to arrive. Your team isn't invited to the QA briefing; that is reserved for the C-staff, namely the COO, CFO, and CIO. (The CEO may also attend if available.) Ok, the meeting is starting now.

Knowledge base: LS08 Validating - QC of SRS by Customer & Vendor + QA of RE.

Activities:

22. Teams [present](#) their Validating method reports. Teams in the audience may utilize whatever inspiration they find from the presenting team in their own document deliverables. (~120 mins.)

Deliverables: There are [three](#): (a) "[Validating QC & QA Rpt](#)", (b) "[Validating QC Prez](#)", and (c) "[Validating QA Prez](#)". Post all into LMS Dropbox folder with the appropriate file names and by the fixed deadline (see *Deliverable details* section). Teacher's assessment will be emailed to PoCs ASAP the following week.

Week 12 User Story & UseCase2.0 | Preliminaries

Role: Customer.

Synopsis: (Your team must re-imagine the requirements capture from the point in time that it has studied the PD and received the CIO's briefing that the SCM for the project will be lightweight and the complimentary RCM is the [User Story](#) method.) Your team is back in the headquarters conference room, waiting for a series of workshops about applying the User Story method. Soon, the CIO arrives with the person that your company has recently contracted to run these workshops. The CIO introduces the person as the [official coach](#) for the [lightweight RCM](#). Then, the CIO informs your team that by the end of the workshops, it will submit a report to the CIO on its progress made in learning the User Story method and show its application on a portion of the PD. The coach begins by having your team open up the drive where the learning material is stored; your team begins reading the material.

Knowledge base: LS10 Overcoming Deficiencies of RE - UserStory is Answer, and LS11 Evolving Familiar & Popular Reqs Capture - UseCase2.0.

Activities:

23. Teams collectively engage in [dialectic session](#) on the following contemplations posed by the teacher with respect to LS10 & 11. (30-45 mins.)
- Both User Story and UseCase2.0 are lightweight RCMs, but they have differences; discuss them?
 - How is the iterative development under UserStory and UseCase2.0 eventually brought to a conclusion?
 - Why is an external coach necessary for UserStory and UseCase2.0 RCMs?
 - After development of new software's core is completed, how should the remaining User Story's be prioritized?
 - Is producing the UseCase graphic model (via UML) useful for effectively writing the UseCase2.0 requirements slices?
 - Also, is publishing the graphic model useful?
 - Is it useful to affiliate individual User Story specifications into aggregates, e.g., epics, themes, features?
24. Teams [apply](#) concepts, hints, examples and ideas from knowledge base and dialectic session to the [undertaking](#) of the document deliverable. (75-90 mins.)

Week 13 User Story & UseCase2.0 | Finalizing

Role: Customer.

Synopsis: (Your team must re-imagine the requirements capture from the point in time that it has studied the PD and received the CIO's briefing that the SCM for the project will be lightweight and the complimentary RCM is the [UseCase2.0](#) method.) Your team is back in the headquarters conference room, waiting for a series of workshops about applying the UseCase2.0 method. Soon, the CIO arrives with the person that your company has recently contracted to run these workshops. The CIO introduces the person as the [official coach](#) for the [lightweight RCM](#). Then, the CIO informs your team that by the end of the workshops, it will submit a report to the CIO on its progress made in learning the UseCase2.0 method and show its application on a portion of the PD. The coach begins by having your team open up the drive where the learning material is stored; your team begins reading the material.

Knowledge base: LS10 Overcoming Deficiencies of RE – UserStory is Answer, and LS11 UseCase2.0 Heavyweight RCM Updated to Lightweight.

Activities:

25. Students pose their questions to teacher concerning any preliminary work on their UserStory & UseCase 2.0 methods, either on behalf of their teams or as individuals. (~30 mins.)
26. Teams continue their [undertaking](#) of the document deliverable. (~90 mins.)

Deliverables: There is [one](#): “[Post-Agile RCM Rpt](#)”. Post it into LMS Dropbox folder with the appropriate file name and by the fixed deadline (see *Deliverable details* section). Teacher’s assessment will be emailed to PoCs ASAP the following week.

Challenging considerations in labtorial conduct

There are four noteworthy aspects to be considered when doing the above labtorials.

1. Role playing! There is no real project, no real tender with customer and vendor, and the teams are not real employee. Nevertheless, the students must assume the [customer](#) role for Eliciting, and the [vendor](#) role for the five remaining learning topics. To that end, the student [must do whatever they can to be the two roles and play act all the notional stuff](#). When you are in the customer role, imagine all the great things that you want in your project, and play-act the desirous nature of a business buying an expensive toy for itself. When you are in the vendor role, imagine you’re a giant killer company making your name for yourself with your leading-edge software. If you don’t do this, not only will you lose out on learning about the different perspectives, but you’ll also lose out on the fun.
2. Finding incongruity with your own work! For the first pair of labtorials, the teams each write a document, the PS. Then, for the second pair, they have to “rip that document apart” in the search for incongruities. This sounds like an impossible task, and for a few it may be, initially. But rest assured, if you play the different roles as fully as possible, every team will be able to find all sorts of incongruities. And if not, the teacher can certainly point out all sorts of deficiencies in the teams’ documents, because most authors are blind to the most obvious of mistakes!
3. It becomes increasingly real after Eliciting! The work in Analyzing involves (a) real Breakdown of a real PS, (b) play-acting the stakeholder engagements, (c) real [prototype](#), (d) imagining some of interpreting, and (e) real posting to real repository. For the last two subprocess, Specifying and Validating, all the work is real.
4. Be honest validators! The team being validated is not being marked; the team doing the validation is. To keep the report results authentic, [no pair of teams will be allowed to validate each other](#). So, feel free to say what you exactly what you see; don’t sugar-coat, and just be professional.

To be clear, the teacher provides substantial feedback during the labtorial, but refrains from giving feedback on the evidentiary documents that students plan to submit. So, the teams must aim their questions at the high-level concepts and themes and not at their specific work.

For example, the teacher will not respond to, “Can you look at our document to see if we are on the right track?”. Instead, the teacher will respond to, “Is there such a condition that there are too many models in an analysis report?” Understand that I will be marking myself if I give feedback or comment upon a deliverable JUST BEFORE it is submitted; that I really want to avoid.

Deliverables

This section overviews the organization of the deliverables, and establishes particular requisites for the production of the deliverable items posted by the students.

Overview

There are five types and fifteen items of standard and assessable work to be delivered in the class project. A summary of the standard types posted by the students is in Figures 4a & b.

Two of the five types are **team** productions that are posted by a designated member of the team into the appropriate **Dropbox** folder. These are:

D type –project documentation, i.e., reports and specifications; total **five** items.

P type –slide deck, i.e., for presentation of reports; total **two** items.

One type is an **individual** effort that is saved by the individual into the appropriate **Quizzes** folder. This is:

Q type – written response quiz; total **two** items.

The final official type is also an **individual** effort that a portion is posted by the individual into the LMS, and the remainder are contemporaneous notes by the teacher about the individual observed efforts. This is:

T type

- notes and templates; posted by individual; total **three** items.
- attendance and presentation performance; notes by teacher; total **three** items.

The single special type and item conducted via email and meetings on demand of a team, and processed as contemporaneous notes by the teacher. (Note: Typically, the degree of collaboration by members of teams is reasonably equal, making a peer evaluation done by all the teams an unhelpful exercise.) This is:

E type – peer evaluation; notes by teacher; total **one** item.

Topic	Code	Deliverable description	LMS	Due
Prior knowledge	T03	Appraisal of prerequisites quiz	Dropbox	Wk01
Product descr (PD)	T05	Individual survey of PD	Dropbox	
Protocol + Reqs gen	T06	Appraisal of learning protocol quiz	Dropbox	Wk02
Eliciting method	D01	Project Specification (PS)	Dropbox	Wk04
Analysing method	D02	Analysing report	Dropbox	Wk06
	P02	Slide deck of Analysing rpt prez	Dropbox	
LS in 1st half	Q01	Quiz 01	Quizzes	Wk07
Specifying method	D03	Sw Reqs Specification (SRS)	Dropbox	Wk09
Validating method	D04	Validating report	Dropbox	Wk11
	P04a	Slide deck of Validating QC prez	Dropbox	
	P04b	Slide deck of Validating QA prez	Dropbox	
UC2.0 & USty	D05	Post-Agile RCM report	Dropbox	Wk13
LS in 2nd half	Q02	Quiz 02	Quizzes	

Fig.4a. Deliverables that students post into LMS per weekly sequence.

Code	Deliverable descr	Due
T02	Presentation performance	Wk06
T04	Presentation performance	Wk11
T07	Attendance (in-class)	Wk13
T08	Participation meaningful	
E01	Peer evaluation (special)	

Fig.4b. Deliverables by teacher per weekly sequence.

Deliverable details

There are five types and fifteen items of standard and assessable work to be delivered in the *class* project. A summary of the standard types posted by the students is in Figures 4a & b.

Specifications

The teams produce **two** specifications, the PS for Eliciting and the SRS for Specifying.

The team chooses the templates that it wishes to follow in the writing of the two specifications. There are many to choose from on the web. However, in keeping with the two roles being played, the templates for PS and SRS must **not be identical**, because they are from imagined different companies doing different industries. Naturally, this is especially crucial for the letterhead and title page. [Google search on “company letterhead for software requirements project” for a plethora of suitable ideas.]

PS. This lists the needs, objectives, constraints, expected features, deadlines and budget for a project to **field a new software in the authoring company**; as part of the tender process, the PS is released to the chosen vendor for their reference, who retain it for traceability and legal reasons. In real-life, the layout of a PS is at the discretion of the company. In this project, the students are free to use the generic layout provided in the Eliciting prez session, or to use any format that seems appropriate. [Google search on “software project specifications”; there are tons of resources available.]

SRS. This contains a **regulated revision** of the PS’ reqrs, plus quality model, tracing matrix, and change mitigation, all wrapped in an international standardized format. (ISO and IEEE provide templates, as do many IT companies.) The SRS is the **only** specification released by the RE. The SRS does not include deadlines and budget; these are held by the vendor’s PM and not for general release to the development team.

Note that the PS and SRS constitute assessable deliverables unto themselves. Do **not** accompany them with a report that explains each specification.

Reports

There are **three** reports, one each for Analyzing, Validating, and Innovative Reqs Capture.

Analysing report. This is a **debriefing report** that explains each activity’s — Breakdown, Clarifying, Interpreting, and Categorizing — objectives, what happened, what went right, what went wrong, its findings, and overall status of the RE operation at this point. To make the content clear, envision that the following scenario has just happened.

The team is relaxing in the office after a couple months of hard work on Analysing, when suddenly the boss bursts in and gruffly announces, “Get off your “cans” and give me a report by Friday on what you jokers did for the last two months! I want to know why I’m paying you, so you’d best tell me

everything, including how the company has benefitted from your la-de-da excursion with one of our customers!”

Validating report. This is a specialized debriefing report, different in **two** ways.

- There are **two** separate documents: ‘quality control’ (QC) findings report, and ‘quality assurance’ (QA) audit report. All members **except one** of the Validating team do the QC; the remaining one member does the QA and only the QA. QC is a search for any overlooked incongruities in the SRS, and the QA determines the extent to which the vendor’s **process** to produce the SRS **met all the contractual and other governing professional obligations and standards**. The two reports are presented **one after the other**, then posted in the LMS **at the same time** (Zip file).
- Teams are **not** Validating themselves; instead, they are validating others. The protocol is, the team **being validated** selects the team that will be its validator. Upon acceptance, all historical record, especially the PD, PS, Analysing report, and SRS of the “validatee” is handed over to the validator. The QA will also need the QC report as soon as it completed.
- As soon as a team has **received acceptance** from its Validating team AND you've supplied all the historical records, **email** teacher immediately with the **name** of the validating team and **confirmation** of the records handover.

Post-Agile RCM report. For this, the teams post a report that demonstrates the application of **User story method** and **UseCase2.0 methodology** respectively to **specify** reqrs for select and limited parts of the **PD**. The report is divided into **two** independent halves, one for the USty generated reqrs and the other for the UC2 generated reqrs. Each half explains how its respective innovative reqrs capture method/methodology is supposed to work, what assumptions had to be made in selecting the target part of the PD, and how the resultant reqrs were actually generated. Keep this report **short**, with just title page, basic ToC, no appendix, and around two-to-four pages maximum each half. There are **two** proprieties to be followed for the content of this report:

- USty and UC2 aren’t **applied to the same parts** of the PD.
- The report **doesn’t make any comparison** between the USty method and UC2 methodology. The team needs only to demonstrate that it has acquired a basic knowledge and skill of how to apply the two innovative reqrs capture method/methodologies.

Presentations

There are **two** prezs, one each for Analyzing and Validating. **All team members** present a fixed portion, or increment, of the whole prez. The assessment of each member’s prez is

upon a demonstrated and convincing display of knowledge, plus a skill for engaging the audience, such as eye contact, boisterous voice, and confidence.

The allowed time for a prez is 12 minutes, so each member talks for about two-to-three minutes. Some overrun of the two-to-three minutes is expected, but an excessive overrun will reduce the mark of the member that committed the overrun.

The prez teams may make last-minute edits to their reports, inspired by whatever transpired during the Q&A, before they are to posted into the LMS.

For the Validating prez, QC always goes first, and QA last (to wrap it all up).

Quiz response

There are two quizzes, one for the content in each half of the trimester. The former is scheduled during the recess week, and the latter for week 13. Each quiz opens online from Monday to Friday of its respective week, is open book, and assigned run-time of 90 mins, after which the quiz closes automatically. Only one attempt at each quiz is allowed.

The students may take each quiz at any time between Monday 0001 hrs and Friday 2359 hours during their respective weeks. However, the student is encouraged not leave it too close to the deadline, after which the quiz is inaccessible.

Each quiz has several open-ended ques. Exceeding 150 words in their response will reduce the students evaluation. All questions are of an equivalent scoring value.

Submitting deliverables

All deliverables are posted into the appropriate folders in DropBox in the LMS (xSite). The deadline for posting is always 1800 hours on the Friday immediately following the finalize and submit labtorial session for each deliverable. The summary of the deliverables and their submission due-weeks are laid out in Figure 4a.

Assessment schema

This section details the schema used to assess the work to be delivered in the class project. There are two primary components in the assessment of the deliverables

- individual component (40%) for each team member, and
- team component (60%) that the entire membership of the team shares.

A summary of the marking schema is in Figure 5. It shows which deliverables are counted in the individual and team components, and what their percentages are in the total grading.

Justifying assessment conduct in this module

For starters, let's establish the terminology being used in this section. **Assessment** is the activity of determining to what extent a student has met the learning outcomes and expectations of the module. **Mark** is the primary output of assessment, a numeric enumeration of goodness for a task in the module, such as submission of a document. **Grade** is the aggregation of all marks, including bonuses or penalties, and moderation post-module. The first two are effected by the **teacher**, and the third is finalized by the **institution**.

Team assmt			Individual assmt		
Code	%	Item	Code	%	Item
D01	11	PS	Q01	10	Midterm quiz
D02	12	Analysing rpt	Q02	10	Final quiz
D03	12	SRS	T02	5	Prez delivery
D04	12	Validating rpt	T03	1	Prereq knowldge
D05	11	Post-Agile rpt	T04	5	Prez delivery
P02	1	Analysing slide	T05	3	Survey PD
P04a	1	QC slide	T06	2	Lrn protocol
P04b		QA slide	T07	2	Attendance
			T08	2	Participation
Total	60			40	

Fig.5. Marking schema for D, P, Q & T type deliverables.

Assessment is the most recognizable component of modern education. Its only truly authentic and feasible function is to provide feedback to students, their teachers, and their institutions, on all-around performance and quality assurance. In ideal conditions, assessment would be well-aligned to the **learning outcomes** and the many **performance criteria** that contribute to their full achievement, such as attendance, breadthwise comprehension of material, creativity, retention, skills acquired, and transfer. (Note: Class **attendance** is a positive causal factor in the student's learning (Burns, 2011; Jones & Sharma, 2021).) Commensurately, the student would be well motivated to excel in the module, and use every assessment as a benchmark for one's degree of comprehension of the material achieved, and a waypoint for "stepping up one's game".

But, the world isn't ideal, and assessments are systemically flawed in practice. Teachers have different approaches to assessment. Assessments are not cross-checked by an

independent agent for the reliability and validity of the assigned mark value. Learning outcomes are usually generalized for manageability, but have the side effect of including criteria that are not true measures of it, such as participation, and ranking within the cohort. There is also significant variance in the assessment between institutions, programs, credentials, and instructors. Even content and instructor variability across multiple sections of the same course can be substantial (Rogers, 2003; PennState, 2021). Then, there is the institution's use the output of assessment (grades) as a final judgment that influences future opportunities for students, teachers and itself. Plus, grades are used for receipt of scholarships or entry into a program or institution. Hence, institutions feel compelled to moderated the grades from the teachers' marks at the course, program and degree levels, and then even further to the GPA.

These result in two negative repercussions for the teacher: grade inflation and the bell-curve. Grade inflation gives students a false sense of achievement in their learning, and blurs the distinction between unusually-talented students and those who are not. The bell-curve adjusts (and lowers for some) marks to fit a predefined distribution, regardless of individual effort, thereby creating anxiety, demotivation, and resentment, plus fostering competition over collaboration. (Recall that [demotivation](#) is antithetical to the learning protocol.)

For students, inadequacies in their assessment, and escalating significance of grades beyond the classroom, reinforces their view that grades are an end in themselves i.e., the primary goal of the education process. Consequently, the attention of many students is only on the learning that will positively affect their grade, as opposed to learning the whole course material. Given these deep-rooted impediments, how can the teacher return to the authentic purpose of [motivating and impassioning the students in learning and engagement](#), and the student return to [focusing on his/her comprehension and retention of course material](#)?

The teacher has enshrined the following [principles](#) for this module's assessment.

1. [Transparency](#): Students will receive clear communication about marking criteria and expectations.
2. [Fairness](#): Students will be marked according to fixed sets of criteria.
3. [Outcome focus](#): Learning outcomes will be adjudicated according to the taxonomy of [demonstrable cognitive capability by Bloom et al](#) (1956) and refined by Anderson and Krathwohl (Eds, 2001). Anything that is not an authentic or presumable part of the established learning outcomes will not be assessed.

Challenging considerations in assessment

There are four noteworthy aspects to be considered when doing the above labtorials.

1. Feedback. The teacher will provide feedback by the following week for common “goodness” or “badness”, plus specific feedback where it seems appropriate. The teacher also endeavours to provide the average score of the marking along with the feedback. Of course, students are encouraged to more feedback on request.
2. Peer evaluation. This is an individual bonus or penalty, the need for which will be identified by students for their team. In other words, it will be done only for and by teams that request it. Know that although it won't by itself cause any student to fail the module, it can either lower or increase the mark of students by a maximum of one grade. If a peer evaluation is to be done, further instructions concerning its conduct will be disseminated. Ensure you rate yourself in addition to the rest of the members of the team. Importantly, take this opportunity to rank each of the team's members for ‘meaningful contribution made to collective project’. The teacher adjudicates how much of a penalty is warranted. It can be any (+ or -) value from 0 to 5.

Adjustments to assessment schema

These are the improvements made by the teacher to the assessment's previous schema.

- [Assessment in sync with learning outcomes](#) – assessment schema is tailored to the new learning outcomes. (Note: Development of both was interleaved.)
- [Design and incorporate modern rubric](#) – innovate standard rubric to increase assessment rigour.
- [Independent assessable items](#) increase number & variety of.
- [Individual marks](#) – increase weightage to over 1/3 of total. (Note: Originally, team collaboration mark was over 2/3 of total.)
- [Learning outcomes](#) – newly created for module and every learning session per Bloom's taxonomy.
- [Stylings' mark](#) – increase [granularity](#) for more rigorous assessment.

Epilogue

Consider the posting of this instruction to be a soft launch. Although great care and thought has gone into this document, it is unlikely that it is without incompleteness and inconsistency. So, the author is grateful for any problems with this instruction to be brought to his attention.

Additionally, the teacher will take the opportunity to check the suitability of these instructions at the start of each of the practice labtorials. This is a good forum for the students to seek any clarifications that they require.

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APPENDIX A

PROJECT DESCRIPTION

In keeping with the explanation of “Description” in Glossary section “High-level documentation in software engineering and SDLC”, a Project Description (PD) is a **set of statements and models** giving the observable or detectable characteristics and real-world attributes of a **Project**, so that its form, functions, and fit in its environment can be imagined, visualized, recognized, and even superficially understood”. For our module, a Project is the development of a new not yet existing **software**, and in the LMS are numerous PDs for such available for the student teams to choose from.

The author of all the PDs is an organization, Nevon Solutions PL. A supply of 280 software projects is offered to the public via an initiative of Nevon Solutions called NevonProjects, that is accessed at, <https://nevonprojects.com/project-ideas/software-project-ideas/>.

The primary concerns vis-à-vis the supply of PDs for this module are each PD is **feasible**, i.e., it can be developed in accordance with the standard SDLC, plus its context of usage (if it were a real-life project) is **neither too simple nor too complex**. After all, the module is at university-level learning of how to devise the requirements for a general-purpose software. The projects’ contexts and industries are sufficiently commonplace and clear-cut that the teams should be equal to the task of applying onto them their newly acquired knowledge and skill of capturing requirements.

Finally, and probably most importantly, all the PDs are **not** specifications. They are deliberately scantily written, without models and codification, so that they provide plenty of room for the students to infuse their own ideas and imagination into the project. In other words, the students will be **transmutating** what exists in the PD, as well as **imagining additional aspects** to fill in any incompleteness’ and incongruities of the project entity.

For example, most PDs don’t stipulate hardware, external systems, or budget and schedule. Definitely, the latter would be stipulated for any real-life software development project; it depends for the others. To ensure that uncertainties in the PD’s feasibility and completeness do not preclude the teams from undertaking the class project, the teacher will advise each team one-on-one on the most effective way to mitigate any perceived issue.

APPENDIX B

BLOOM'S TAXONOMY OF LEARNING GOALS

APPENDIX C

EFFECTIVE WRITING OF REQUIREMENTS FOR ELICITING METHOD

From idea to specification

The Eliciting Method's purpose is to [draw forth](#) the requirements for some new not yet existing software and then document them using the stylized form called [requirements clauses](#) into a foundational document called the [Project Specification \(PS\)](#). More succinctly, it is designing [original](#) software requirements [from an idea to a specification](#). This appendix explains how to write these clauses [effectively](#) despite the challenge inherent of achieving originality along with quality.

“Creative ideas for a starving artist” (paraphrasing). Designing something that has never existed before, is a journey of originality and invention of both product and its production process. Assume you’ve passed the right of passage — is it even important to undertake this challenge — the most crucial uncertainty is, where do you start? Perhaps it is, learn the mistakes that that were made before and don’t replicate them. Then, figure out what it is you don’t know, but need to know, and then find a way to know it. Apply your own ideas and knowledge of what works, experiment, strategize, establish your own approach and tactics and follow your own instincts. You’re going to have to make the method or process up all by yourself. You’re going to have to be imaginative, innovative and even courageous. But how?! (by [tropicaltheartist](#), [February 27, 2012](#))

Always remind yourself that the software being developed cannot be a replica of what the recipient company has now. If that were the case, that company would never embark on a such a [costly and risky endeavour](#) as development. Instead, it would merely [overhaul](#) the existing software, or [replace](#) it with a commercial off the shelf (COTS) software that appears to be “next level”.

Fortunately, when designing something that has never existed before, rarely is there absolutely nothing to go on. (Let’s reject the condition of discovery, where it does exist but has gone unnoticed until now.) There is likely no stipulation that the new thing has to be [totally new](#), so you could leverage parts of existing things or methods that are proven sound upon which to start your designing. Also, there may exist a moderately adaptable [precedent approach](#) for designing something “new out of the notion of” some [tangible predecessor](#).

Still, the certainty remains that a project to design something original would necessitate climbing a [learning curve](#) before the something being developed achieves an appropriate facsimile of the idea that spawned the development in the first place

The steepness of the learning curve depends on whether a precedent approach [exists](#) plus is imbued with the [right lessons](#). For the companies that have a precedent approach, very likely it's emerged from the experiences of the several software requirements projects "under their belt". Yet, even the most adaptive and reliable precedent approach can't be expected to be effective for all of the myriad kinds of software needed in the market. The road to originality and invention is not be a smooth one.

Fortunately, your *team* projects have been engineered to give you a fair chance at doing a credible job of drawing forth the software requirements. You have an idea of what the software might be from the PD you selected. (See Appendix A.) The software described by the PD is [neither too simple nor too complex](#) and seems appropriate for university-level learning. You have, from a previous module, a beginner's idea on how to write requirements clauses. Reasonably, you can leverage that as having done a previous requirements project. Finally, you will forthwith be provided with an adaptable precedent approach for effectively writing the requirements clauses.

How to write requirements effectively

The following are two aids in writing the requirements effectively in Eliciting method: Magnificent Seven quality model and Requirements Tree. Let's begin with the Magnificent Seven quality model (Figure 6). As the name suggests, there are seven quality elements in the model. One prescribes the quality of the set of the requirements clauses, while the remaining six are for individual clauses. Let's start with the latter.

1. Correct. Each clause must state the requirement [accurately](#) (not as guess, maybe, or approximation) and [don't contradict](#) any other clause ("Two men say they're Jesus, one of them must be wrong" – Dire Straits). In particular, avoid the use of repeating and infinite fractions and irrational numbers as boundaries or in equations. (Note. A side effect of decimal-to-digital conversion is that approximating repeating fractions over time, such as one-third, can cause the build-up of a significant error. This was the root issue in the Patriot Missile failure.)
2. Fully distinct. Do not [repeat](#) an entire clause (be [unique](#)). Do not repeat any part of any other clause ([no overlap](#)). If a requirement is expected to give rise to numerous separate clauses upon its creation, then write them as a list, where each item is itself unique and not overlapping with any other. In the case that several items that are distinct requirements in their own right, but which must work in conjunction or as a

sequence after they are created, avoid making them into a list. (Note. Realistically, every requirement in the software will have some dependency upon others. Only show those that are not normal or intuitive.)

3. Ranked. Enumerate clauses to indicate relative sequencing (for process) and priority (for importance of implementation or quality). There may be one ranking across all clauses, or several different rankings for several different criteria across some of the clauses.

4. Realistic. Each clause states a requirement that can be realizable or satisfiable by virtue of existing and accessible technology and industrial know-how. Since the access to different technologies is not uniform

across all possible realizers of the requirements, it is acceptable that seem unrealistic to some. The main contention of realism is whether it is unrealistic to **all possible realizers**. The author of the requirements might think it is only unrealistic for some, and use that as a way to subtly elicit a response from a subset of the possible realizers. In this case, then the requirement must be ranked/prioritized as **optional**.

5. Unambiguous. Every clause's meaning must be **emphatically singular**. To that end, the expression of each clause must be clear, precise, and definitive so that the understanding that is intended to be gleaned comes through easily.

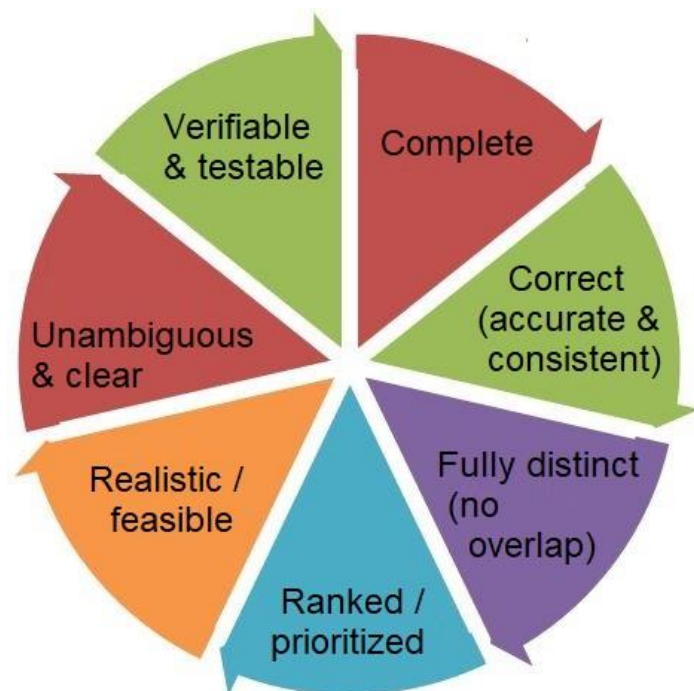


Fig. 6. Magnificent Seven quality model for requirements clauses.

The analogy of a **jigsaw puzzle** can best explain the last two elements. There are a fixed number of pieces, and every piece has a set of two-to-four other pieces that connect to it. The puzzle is constructed by pressing each piece into its rightful place in another piece. However, each piece is engineered so that there is one place within the whole of the puzzle that it can truly fit into to (without damaging other pieces). The puzzle is finished when every piece is in place and none unplaced remain. The puzzle has two built-in two aids to facilitate its finish: a picture of the puzzle of which every piece displays a small part of, and a set of easy-to-recognize border pieces that, when connected, form the outer frame of the puzzle. In other words, all connections are verifiable, and completeness is demonstrable.

6. Verifiable. Each clause must be **necessary** (it exists so that another requirement can be realized or satisfied) and/or **sufficient** (it can realize or satisfy its requirement on its own). There is a derivative of verifiability called **testability**, where the requirement clause is demonstrated with a prototype or simulator. Otherwise, verifiability is determined through inspection, walkthroughs, thought-experiment, outcome analysis, or role-play as user. (Note: the underlying principle that supports verifiability is falsifiability: the clause must have content and details enough to be demonstrably true or false.)
7. Complete. All requirements for the new software are accounted for in the set of clauses. In a perfect world, the **stakeholders** may be mustered to validate that every single one of their requirements are present in the final set of clauses. Alternatives are possible, such as other **informal** methods — their utility depends on themselves being complete — and one **formal** (Carson, 2004) — too complicated to use in this module — but we will not use any. Instead, the teacher will provide his own method.

Effective writing of completeness

The teacher's method is the **Requirement Tree**, a simple 'has-A' or composition hierarchy for breaking down high-level and imprecise requirements into more granular and detailed ones. The decomposition breakdown or drill-down, is easy to read and understand. Stakeholders and realizers can easily visual the big picture plus the dependencies between requirements. Most software engineers engaged in capturing requirements will create their own Requirements Tree; for this module, mine (see Figure 7) will serve as a suitable inceptive precedent approach. You may change the class names and add classes; just **don't subtract** from the model.

The following list details the description and examples of the realization of each class, starting at the root of the hierarchy.

Level 0 System

Description: High-level classification of scale, scope, and type of SDLC project.

Realization e.g.: **Architecture**

(Application/standalone program, component-based/ micro-services; client-server/full stack/front-end/back-end;

distributed/serverless/SOA/SAAS), **Regulations** (universal single or class several), and

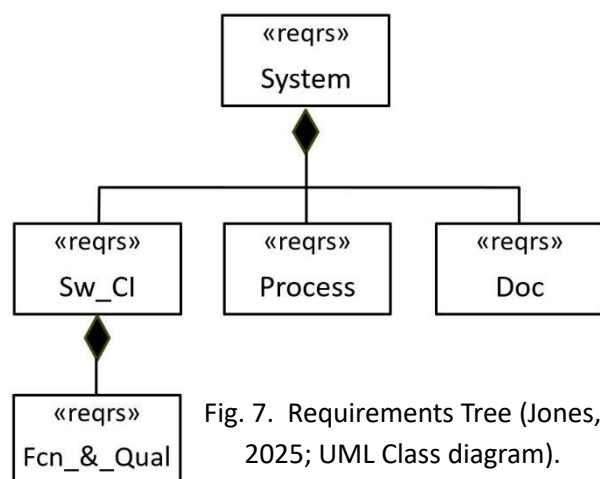


Fig. 7. Requirements Tree (Jones, 2025; UML Class diagram).

Assumptions (high-level only). (Note. Developing new not yet existing hardware is not in the scope of this module. If hardware is included in the *team* project scope, it must be **COTS** only.)

Level 1 Process

Description: What directly & indirectly done by people to enact and support the commerce & viability of the owning organization.

Realization e.g.: Its relation to processes either/both **business** process (how the new software is going to be used, such as manufacturing, product or service provider, in supply chain, warehousing, health and wellness, research, non-profit service) or **non-business** process (supports the business with platform process, data process, security, interface protocols, technical facilitation, related infrastructure) .

Level 1 Doc

Description: Represents the detail of content, formatting, ontology, organization, and terminology.

Realization e.g.: Corporate promotion material, data dictionary, helpdesk SOPs, operating instructions & policies, maintenance protocols, online help, models, queries, reference/handbooks, reports, requirements traceability matrix (RTM), screen capture macros, session logs, software tools & utilities for system info, training, and user guides.

Level 1 Software Configuration Item

Description: Any software that is to be registered as a configuration item in the inventory of the recipient company's assets, meaning they have real monetary value. Typically, these are to be listed in the final specification of the requirements as deliverables, e.g., Google Maps is not a deliverable, but an interface to it for mission planning is.

Realization e.g.: Main program; its database management software; its platform software; its direct peripherals including closed circuit software; external system software interface; indirect communications software including network, audio IO, and visual IO. (Note. Any **COTS** software that is paid for and delivered as part of the project is included in this class as a configuration item.)

Level 2 Function and Quality

Description: **Functional** (what it does) and **quality** a.k.a., non-functional (what it is) elements of each of the Level 1 softwares that are new and not yet existing.

Realization e.g.: Create-Read-Update-Delete (CRUD), Representational state transfer (REST), API/external system interface contracting, messaging (follow, post), SQL, UI/UX,

and quality attributes including access control, analytics, authentication, consistency of data, customizing, history, maintenance, metrics, performance, reliability, and security.

Importantly, this hierarchy is not based on inheritance. There are no common features passed from the root to Level 1 then to Level 2, and so on. Instead, the tree is about containment. The requirements of the system contain the requirements of software, processes related to the new software configuration items, and documents related to the new software. Each of the new software configuration items contain the requirements of functions and quality.

With the right experience, a software engineer can build their own Requirements Tree that serves as their precedent approach. In particular, the key the approach is getting the Level 0 requirements correct. In the analogy of the jigsaw puzzle, Level 0 acts as both the picture and outer frame of the puzzle. Once this is done well, the rest of the requirements should fall into place like pieces of the puzzle.

APPENDIX D

QC AND QA CONDUCT INFORMATION